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A hybrid ANP and machine learning framework for contractor health safety environment and quality performance evaluation

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Abstract

Contractor-related deficiencies remain a critical challenge in workplace health and safety management within high-risk industries, particularly in the oil and gas sector, where heterogeneous safety practices and fragmented oversight mechanisms undermine effective risk control. Existing contractor evaluation approaches often rely on checklist-based or lagging indicators, offering limited ability to capture interdependencies among safety dimensions or to differentiate contractor performance in a meaningful and decision-relevant manner. This study develops and validates the Contractor Health, Safety, Environment, and Quality Performance Index (CHPI) as an integrated, evidence-based framework for systematic contractor performance evaluation. The study adopted a multi-method approach integrating expert judgment and archival compliance data. Health and safety indicators were identified through literature review and expert consultation, refined using Content Validity Index assessment, and weighted using a Fuzzy Analytic Network Process to capture interdependencies. The resulting weights were combined with standardized contractor records to compute CHPI scores. Robustness was confirmed through sensitivity analysis demonstrating stable contractor rankings, while a Random Forest-based analysis was used as a complementary validation to assess alignment between expert-based weights and data-driven importance. The results show that the CHPI enables nuanced differentiation of contractor performance, supports pre-contract screening and targeted intervention strategies, and enhances transparency in performance-based regulation. By integrating interdependency-aware weighting with empirical validation, the CHPI provides a scalable and adaptive decision-support tool that can strengthen contractor governance, improve safety performance monitoring, and support societal progress through more accountable risk management in high-risk industrial environments.

Keywords Contractor safety performance, Analytic network process, Machine learning, Performance index, Proactive approach

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1 Introduction

1.1 Evolution of HSE performance indicators

The oil and gas industry are widely recognized as one of the most hazardous sectors due to its dependence on complex technical systems, high-pressure processes, and the use of flammable or toxic materials [1]. Accidents in this sector often have severe human, environmental, and economic consequences, making robust workplace health, safety and environment (WHSE) management not merely a regulatory obligation but also a strategic necessity for organizational resilience [2]. Within this context, contractors play a decisive role in determining overall WHSE performance [3]. Nevertheless, empirical studies have shown that contractor compliance varies widely due to differences in safety culture, training quality, and enforcement mechanisms [4]. This variability creates a strong need for more systematic, data-driven approaches to monitoring and improving contractor WHSE performance. Additionally, contractor-related issues contribute to WHSE failures through multiple organizational and operational pathways. Contractors often exhibit heterogeneous safety cultures, varying levels of training, and inconsistent adherence to host organization procedures, which can weaken control mechanisms at the site level [5]. High workforce turnover, fragmented communication channels, and misalignment between contractual pressures and safety priorities further exacerbate these risks [6]. In the oil and gas industry, such vulnerabilities are amplified due to the inherent presence of hazardous materials, high-pressure systems, complex process technologies, and tightly coupled operations where minor deviations can escalate into major incidents [7]. As a result, deficiencies in contractor oversight, competency management, and real-time monitoring play a disproportionate role in WHSE failures, making contractor performance evaluation a critical component of risk management in high-risk oil and gas environments. Against this backdrop, improving contractor WHSE performance requires not only effective governance mechanisms but also robust performance measurement systems, making the evolution of WHSE performance indicators a central concern in high-risk industries such as oil and gas.

Over the past two decades, numerous studies have attempted to develop frameworks for measuring and monitoring WHSE performance in high-risk industries. Early approaches relied heavily on lagging indicators such as lost-time injury frequency rates (LTIFR) [8] and total recordable incident rates (TRIR) [9]. While these metrics remain useful for compliance verification, they offer limited predictive insight and cannot adequately identify emerging risks [10] before incidents occur. In response to these limitations, researchers have increasingly emphasized the role of leading indicators, proactive measures [11] that signal deteriorating safety conditions before they result in accidents. Examples include hazard identification rates, frequency of safety audits, participation in safety training, and the effectiveness of emergency preparedness drills [12]. In the Iranian oil and gas sector, Sarkheil and Rahbari [13] and Amir-Heidari et al. [14] demonstrated that integrating leading indicators into WHSE management systems (HSE-MS) can enhance sustainability and long-term safety performance. Azadeh et al. [15] used multivariate analysis to continuously assess integrated WHSE and maintenance systems in gas transmission units, showing how statistical models can identify performance improvement opportunities. However, despite these advances, many indicators set remain context-specific [16] and lack the flexibility for cross-sector or cross-country

application [17]. This points to a need for indicator frameworks that are both adaptable to local conditions and aligned with international best practices.

Despite the growing body of research on WHSE performance indicators in high-risk industries, existing studies reveal several unresolved limitations that motivate the present research. Prior efforts have predominantly focused on developing context-specific indicator sets or evaluating isolated safety dimensions within single organizational or operational settings. For example, studies conducted in Iranian oil and gas and chemical industries have demonstrated the value of active and leading indicators in improving safety outcomes through administrative interventions, integrated management systems, and conceptual performance models [18–21]. While these contributions have advanced understanding of WHSE measurement at the organizational level, they often rely on static indicator structures, limited interdependency modelling, and descriptive or sector-bound validation approaches. Moreover, contractor safety performance, characterized by fragmented responsibilities, heterogeneous safety cultures, and dynamic operational conditions, remains underexplored within these frameworks. These gaps highlight the need for a more integrative, empirically validated approach that captures the systemic interactions among WHSE indicators and supports consistent contractor performance evaluation in complex, high-risk environments. Addressing this need provides the primary motivation for developing the CHPI framework proposed in this study. In this context, the Contractor Health, Safety, Environment, and Quality Performance Index (CHPI) is conceptualized as an integrated and composite framework for evaluating contractor WHSE performance in high-risk industrial settings. The CHPI systematically aggregates validated safety indicators across behavioural, organizational, and operational domains, explicitly accounting for interdependencies among indicators through network-based weighting. By combining expert-informed prioritization with empirical performance data and predictive validation, the CHPI enables consistent benchmarking, risk-based contractor differentiation, and evidence-driven decision-making. This unified definition establishes CHPI as a performance-oriented evaluation model rather than a descriptive indicator set, providing a coherent foundation for the methodological approach adopted in this study.

1.2 Contractor HSEQ performance in relation to organizational safety governance

Contractors play a central role in shaping organizational HSEQ performance in high-risk industries, as a substantial proportion of operational activities are outsourced to external firms. Although contractors operate under the oversight of the host organization, their safety practices, workforce competency, and environmental controls often vary considerably, creating systemic vulnerabilities within organizational HSEQ governance [5]. Deficiencies in contractor performance can directly undermine organizational safety objectives, increase incident likelihood, and expose organizations to regulatory, reputational, and financial risks [3]. In contractor-intensive sectors such as oil and gas, the organizational responsibility for HSEQ outcomes extends beyond internal systems to include the effective selection, monitoring, and control of contractor performance [22].

Traditional organizational HSEQ management systems frequently rely on compliance-based checklists or minimum qualification requirements, which provide limited insight into how contractor practices interact with organizational controls and risk management mechanisms. This disconnect reduces the organization's ability to proactively identify

underperforming contractors, prioritize interventions, and align contractor behaviour with organizational safety expectations [23]. Accordingly, effective organizational HSEQ governance requires structured, transparent, and performance-oriented mechanisms capable of capturing the multidimensional and interdependent nature of contractor activities [24]. An integrated CHPI, aligned with organizational safety objectives and validated using empirical data, offers a practical pathway for strengthening contractor oversight, supporting evidence-based decision-making, and enhancing overall organizational HSEQ performance. The CHPI framework proposed in this study is developed within this organizational context, explicitly addressing the need to evaluate contractor HSEQ performance as an integral component of organizational risk management and safety governance.

1.3 Advanced data-driven and hybrid approaches in contractor WHSE performance evaluation

Given the multifaceted nature of WHSE performance, researchers have increasingly turned to multi-criteria decision-making (MCDM) methods [25] to evaluate and prioritize safety indicators. Among these, the ANP [26] has gained attention for its ability to capture feedback loops and interdependencies that are often oversimplified in hierarchical models such as the Analytic Hierarchy Process (AHP) [27]. Applications of ANP in high-risk industries have included contractor selection models [28], risk prioritization frameworks [29], and safety system evaluations [30], for example, applied Fuzzy ANP to prioritize WHSE performance indicators in large industrial organizations [31], demonstrating that competency-related measures often influence broader domains such as risk control and emergency preparedness. However, most of these studies remain confined to pre-contract evaluation or narrow organizational settings and rarely extend to continuous contractor performance monitoring during active project phases. Moreover, empirical validation of ANP-derived weights using real operational data remains limited.

In parallel, advances in statistical modelling and machine learning have expanded the use of data-driven techniques for forecasting complex and high-uncertainty phenomena. Gul et al. [32] demonstrated the superiority of hybrid statistical-machine learning models in predicting COVID-19 mortality trends, showing improved predictive accuracy across MAE, RMSE, and MAPE metrics. Beyond the pandemic context, such findings highlight the broader applicability of ML-based approaches in safety-critical domains characterized by uncertainty and nonlinear system behaviour [33]. Complementary evidence from occupational safety research indicates that machine learning techniques can enhance decision-making by identifying latent patterns in operational data [34] Similarly, Sarwar et al. [35] demonstrated that intelligent feature selection integrated with machine learning significantly improves predictive accuracy and interpretability in complex risk assessment contexts. In construction and process safety, algorithms such as Random Forests [36] and Bayesian networks [37], and other ML techniques have been applied to accident prediction [38], hazard identification [39], and proactive safety management [40].

Despite these advances, the two research streams, structured MCDM-based evaluation and data-driven prediction—have largely evolved in isolation. Framework-based contractor evaluation models typically rely on expert judgment and predefined scoring structures without empirically validating indicator relevance against operational

performance data [41, 42]. Conversely, machine learning studies often focus on outcome prediction and pattern recognition without embedding indicators within a theory-informed and interdependency-aware structure [43–45].

Despite substantial advances in safety performance assessment, existing contractor evaluation approaches continue to exhibit important conceptual and methodological limitations. Framework-based models frequently produce static, checklist-driven scores without demonstrating predictive relevance, while machine learning-based studies often prioritize statistical accuracy without embedding indicators within a structured and theory-informed decision architecture. This methodological disconnect leaves regulators and asset owners without a defensible basis for prioritizing safety interventions or justifying contractor differentiation under real operational conditions. A robust evaluation framework therefore requires simultaneous theoretical grounding, interdependency-aware modelling, and empirical validation. Addressing these gaps, this study develops and validates the CHPI as an integrated and empirically supported evaluation framework. The novelty of this research lies in systematically combining content-validated indicator selection, interdependency modelling through Fuzzy ANP, and machine learning-based feature importance analysis within a unified decision structure. Unlike prior contractor safety indices that rely exclusively on expert judgment or standalone predictive analytics, the proposed CHPI reconciles structural expert-derived priorities with operational empirical relevance. This dual-perspective validation mechanism enhances methodological robustness, interpretability, and regulatory applicability, positioning the CHPI as a scalable and decision-relevant advancement in contractor WHSE performance evaluation.

2 Integrated socio technical safety governance framework

The development of the CHPI is grounded in what we define as an Integrated Socio Technical Safety Governance Framework, which synthesizes Safety Performance Theory, Socio Technical Systems Theory, and multi criteria decision principles to structurally model contractor WHSE performance [46], supported by principles of MCDM (Fig. 1). Together, these frameworks provide a conceptual foundation for understanding and evaluating contractor safety performance in high-risk industrial environments such as the oil and gas sector.

Safety Performance Theory [47] posits that safety outcomes are the result of the dynamic interaction between organizational systems, individual behaviours, and environmental conditions. Within contractor-heavy operations, safety performance is shaped not only by compliance with technical standards [48] but also by proactive measures such as hazard identification, competency development, and effective emergency preparedness [49]. This theoretical lens underscores the importance of incorporating both leading indicators and lagging indicators into performance assessment tools. The CHPI operationalizes this principle by integrating a validated set of indicators that capture preventive, behavioural, and outcome-based dimensions of safety.

Socio-Technical Systems Theory conceptualizes organizations as interdependent networks of human and technical subsystems, where safety emerges from the coordinated functioning of these components [50]. In the context of contractor management, this means that factors such as communication quality, leadership engagement, and documentation processes [51] are closely linked to technical controls like equipment

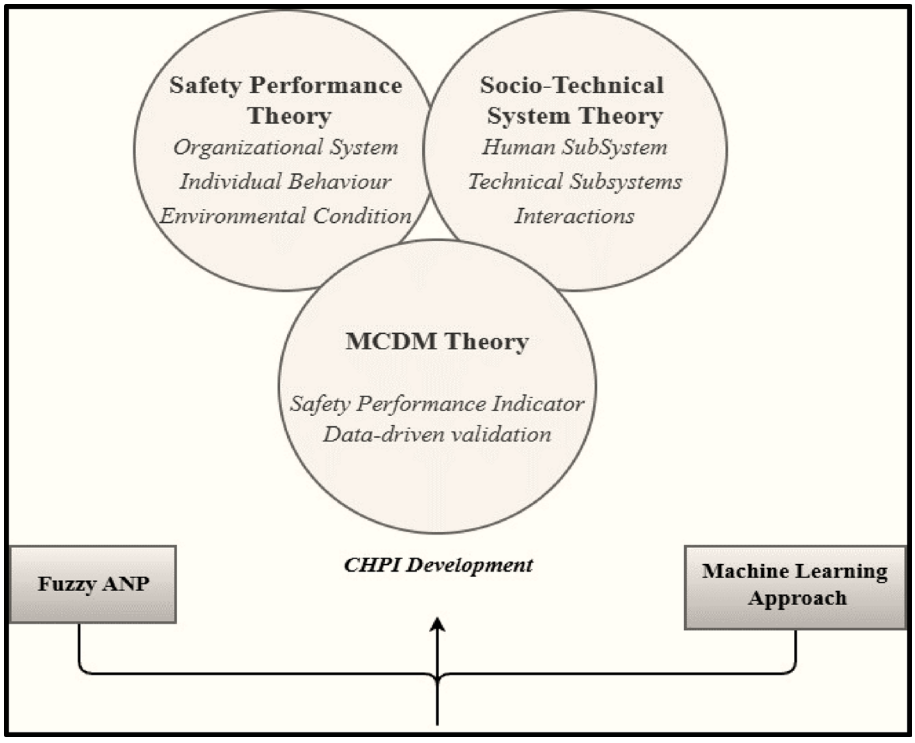


Fig. 1 Conceptual architecture of the integrated socio technical safety governance framework underpinning CHPI development

maintenance, hazard monitoring, and emergency systems. The CHPI framework reflects this systemic interdependence by using the ANP to model feedback loops and cross-domain influences among WHSE indicators, enabling a more realistic representation of contractor performance dynamics compared to linear or hierarchical models.

Finally, the integration of MCDM principles with data-driven validation [52] responds to recent calls for more robust and predictive safety assessment models. The CHPI applies Fuzzy ANP to capture expert judgments under uncertainty, while machine learning serves as a secondary validation layer to evaluate the predictive relevance of each indicator. This hybrid approach aligns with the trend towards intelligent [53], adaptive safety management systems [54] capable of continuous monitoring and early risk detection. In sum, the theoretical foundation of CHPI bridges behavioural, organizational, and technical dimensions of safety performance, aligns with established safety science frameworks, and incorporates advanced analytical methods to enhance both the validity and operational applicability of contractor performance evaluation.

3 Methods

3.1 Research design

This chapter outlines the methodological framework adopted to develop and validate the CHPI. It describes the overall research design, indicator identification and validation process, data collection procedures, analytical techniques, and validation strategy. The methodological approach was structured to ensure conceptual rigor, empirical robustness, and alignment with the study's objective of integrating expert-based multi-criteria decision-making with data-driven machine learning validation.

This study employed a cross-sectional, multi-method design to evaluate the WHSE performance of 72 contractors operating under the East Azerbaijan Gas Company (EAGC) during the 2022–2023 fiscal years. The methodological framework integrated expert judgment, archival performance data, and advanced modelling to develop and validate the CHPI (Fig. 2). The process involved three sequential stages: prioritizing interdependent WHSE indicators using the Fuzzy Analytic Network Process (F-ANP) to obtain domain-specific and global weights; standardizing archival contractor compliance records and combining them with these weights to compute composite CHPI scores; and applying a Random Forest regression model to assess the predictive validity of the CHPI framework.

The selection of Fuzzy ANP over alternative hybrid approaches such as DEMATEL-ANP or BWM-ANP was methodologically driven. While DEMATEL-ANP is often employed to distinguish causal from effect indicators and quantify directional influence strength, the primary objective of this study was not causal mapping but interdependency-aware priority derivation within a structurally integrated network. Similarly, although BWM-based extensions reduce the number of pairwise comparisons, they are more suitable for relatively hierarchical or partially connected structures and are less frequently applied to densely interlinked feedback systems. Given the multi-cluster, reciprocal, and feedback-driven architecture of contractor WHSE indicators in this study, Fuzzy ANP provided a more appropriate modelling framework for capturing bidirectional dependencies under epistemic uncertainty. The fuzzy extension further enhanced robustness by accommodating linguistic variability in expert judgments.

3.2 Identification of candidate WHSE indicators

The development of a comprehensive pool of measures tailored to contractor WHSE performance was guided by a structured indicator-scoping process that integrated

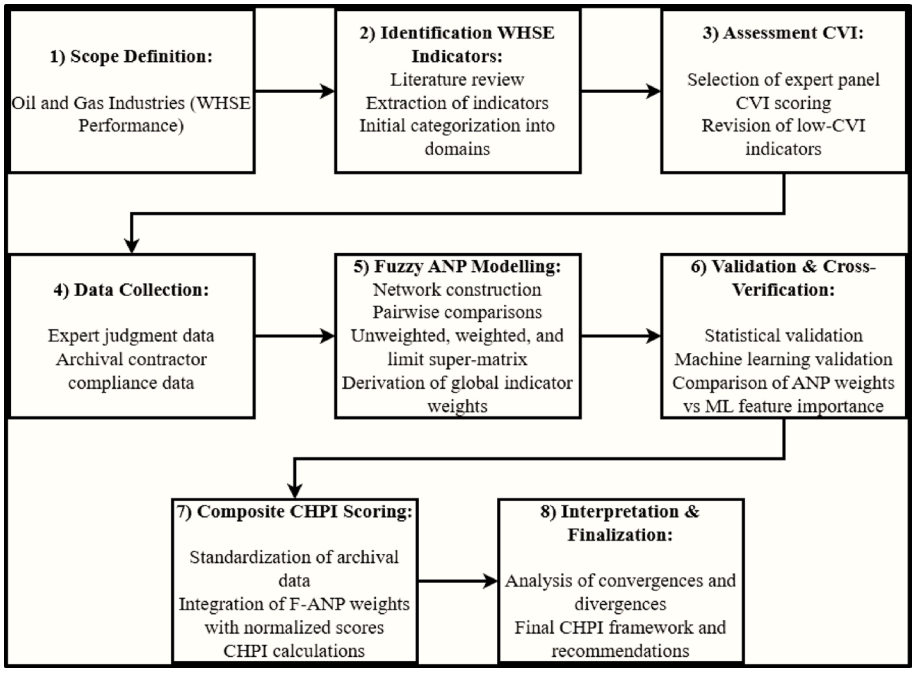


Fig. 2 Structured workflow of the CHPI development and validation process

multiple sources of evidence. This process drew on international WHSE guidelines issued by recognized bodies such as the Occupational Safety and Health Administration (OSHA), the International Labour Organization (ILO), and the American Petroleum Institute (API), alongside peer-reviewed academic literature on safety performance metrics in high-risk industries, and internal policy documents and compliance protocols from the EAGC and the National Iranian Gas Company (NIGC). Both English and Persian materials published between 2005 and 2024 were examined, with selection criteria focusing on indicator relevance, field observability, and compatibility with regional regulatory requirements. Literature-based indicators were identified through targeted keyword searches, including terms such as "contractor WHSE indicators," "occupational risk metrics," and "compliance performance," while internal documentation contributed additional measures rooted in local operational realities; these were retained only when they demonstrated both practical applicability and conceptual consistency with established safety frameworks. The outcome of this scoping exercise was an initial set of 84 candidate indicators, each corresponding to a distinct WHSE construct, which were subsequently organized into seven preliminary domains thereby establishing a coherent foundation for systematic expert review and subsequent ANP modelling.

3.3 Data collection

Experts were selected using purposive sampling based on predefined eligibility criteria to ensure domain relevance and professional credibility. Inclusion criteria comprised: (i) a minimum of 8 years of professional experience in WHSE management, occupational safety, environmental health, or process risk assessment; (ii) at least a bachelor's degree in a relevant field such as occupational health, safety engineering, environmental engineering, or industrial management; (iii) direct involvement in contractor oversight, HSE auditing, or safety system implementation in high-risk industries; and (iv) demonstrated familiarity with WHSE management systems and performance evaluation frameworks. These criteria were applied to ensure that expert judgments reflected both theoretical knowledge and practical, field-based experience.

To ensure the conceptual rigor, clarity, and operational relevance of the proposed contractor-level WHSE performance indicators, a structured content validity assessment was conducted prior to their inclusion in the ANP framework. A panel of 14 subject-matter experts, averaging 10.8 years of professional experience (Table 1), evaluated 84 candidate indicators across three dimensions: relevance to contractor WHSE performance, clarity of definition, and feasibility of implementation and monitoring. The panel size is consistent with methodological recommendations for ANP-based decision modelling in complex systems, where expert panels commonly range between 8 and 20 participants, particularly when members possess substantial domain-specific expertise. Given the cognitive demands associated with interdependent pairwise comparisons in

Table 1 Profile of experts involved in content validity assessment

Expert role	Number of experts (*n*)	Mean years of experience
Project-Level WHSE Managers	4	11.4
Senior Safety Officers	3	10.2
Environmental Health Professionals	3	11.0
Process Risk Assessors	2	9.7
Certified HSE Auditors	2	12.1
Total	14	10.8

ANP structures, emphasis is typically placed on depth and relevance of expertise rather than large sample size.

Using a 9-point rating scale and Lawshe's method, CVRs were calculated, with 0.51 set as the minimum acceptable value for the 14-member panel. Of the 84 indicators, 81 met or exceeded this threshold and were retained; the remaining three were revised based on expert feedback and re-evaluated. Following indicator validation, a dual-source data collection strategy was adopted to generate the empirical dataset required for ANP weighting and CHPI computation. The same expert panel provided pairwise comparisons using Saaty's 1–9 scale, covering both intra-cluster and inter-cluster relationships among the 43 validated sub-indicators. Responses were obtained through in-person meetings or electronic questionnaires and aggregated using the geometric mean method to construct fuzzy pairwise comparison matrices. Consistency Ratios (CRs) were computed in Super Decisions (v3.2) to verify logical coherence, with matrices exceeding 0.10 flagged for revision. Two refinement rounds were conducted, resulting in a low average CR of 0.054, indicating strong judgment coherence and further supporting the robustness of the derived weights.

The second stream involved collecting empirical compliance records from 26 contractor organizations operating under the EAGC in infrastructure, pipeline maintenance, and industrial service projects. Archival documentation was systematically reviewed, with each record mapped to its relevant WHSE sub-indicator and scored on a standardized 0–100 scale. Binary indicators were scored 0 or 100, while quantitative measures were normalized via linear transformation. Missing entries were scored as zero unless clarified via follow-up with site personnel. Two independent analysts assessed the records, resolving discrepancies by consensus. The resulting normalized compliance scores were integrated with the F-ANP-derived weights to compute CHPI scores for each contractor. This dataset was also used to train a Random Forest regression model, supporting the evaluation of the CHPI framework's predictive validity, and reinforcing its empirical robustness.

3.4 Indicator development and reduction process

The development of the CHPI indicators followed a structured, multi-stage refinement process to ensure content validity, relevance, and operational applicability (Fig. 3). Initially, a comprehensive pool of 84 candidate indicators was compiled based on an extensive review of prior contractor safety evaluation studies and WHSE standards. In the first screening stage, redundancy and semantic overlap were addressed through expert review, resulting in the removal of three overlapping indicators and yielding a refined set of 81 indicators. In the second stage, content validity was assessed using the CVR and CVI. A CVR threshold of 0.51 was applied in accordance with Lawshe's table for a panel of 14 experts, ensuring that retained indicators were considered essential by most experts. Subsequently, indicators meeting a CVI threshold of 0.78 were retained to confirm clarity, relevance, and representativeness. This dual filtering process reduced the indicator set to 43 validated indicators. In the final stage, the retained indicators were conceptually grouped based on functional similarity and system-level interactions, resulting in seven thematic clusters: Awareness and Competency, Risk Management, Emergency Preparedness, Environmental Health Management, Documentation, Monitoring and Audit, and WHSE Management Systems. This seven-cluster structure was

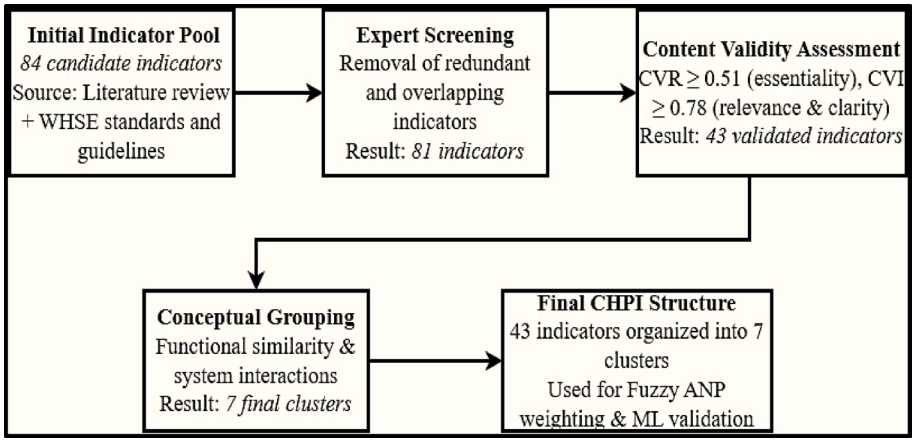


Fig. 3 Indicator refinement and structural consolidation process for CHPI construction

consistently adopted throughout the ANP modelling and empirical validation phases and represents the final organizational structure of the CHPI framework.

3.5 ANP network construction and weight derivation

Given the interdependent nature of contractor WHSE indicators, a hierarchical method such as AHP was considered insufficient. Therefore, ANP was adopted to capture feedback and mutual influences among indicators. Based on the validated indicators, the decision network was organized into six clusters: Awareness and Competency, Documentation, Emergency Preparedness, Environmental Health Management, WHSE Management Systems, and Monitoring and Audit, collectively representing the key organizational, procedural, and operational dimensions of contractor safety performance. The evaluation of contractor WHSE performance involves multiple criteria that are inherently interdependent and dynamically linked rather than hierarchically independent. In such contexts, traditional hierarchical methods such as the AHP, which assume unidirectional relationships and independence among criteria, may oversimplify the underlying safety structure [30]. Also, the ANP is specifically designed to capture these complex interrelationships by allowing feedback loops and interdependencies among criteria [55]. Given the strong interdependencies among contractor WHSE indicators, a hierarchical decision structure was deemed insufficient; therefore, ANP was adopted to explicitly model feedback and mutual influence among safety dimensions.

To quantify indicator interdependencies, a structured pairwise comparison questionnaire was developed in accordance with the ANP framework, capturing both intra- and inter-cluster influences (Fig. 4). Expert judgments were processed using Super Decisions v3.2, with all consistency ratios remaining below the acceptable threshold of 0.10, confirming the reliability of the comparisons. ANP calculations were then performed to derive global priority weights through successive construction of the unweighted, weighted, and limit super-matrices. The results (Table 2) indicate that Awareness and Competency held the highest global priority (0.358), followed by WHSE Management Systems (0.148), while Documentation (0.086) and Environmental Health Management (0.093) exhibited comparatively lower influence on contractor WHSE performance differentiation.

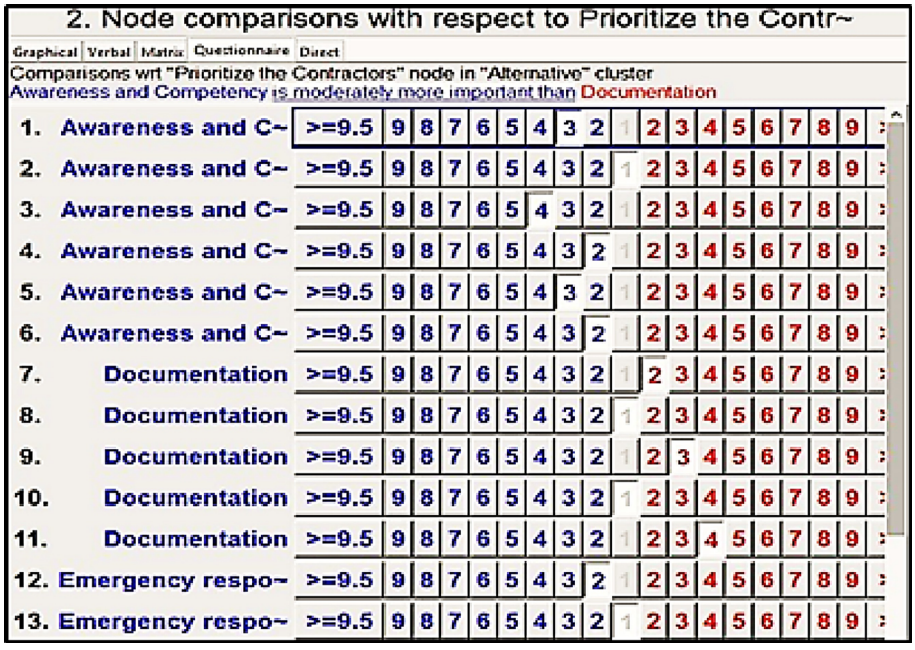


Fig. 4 Pairwise comparison by super decision software

Table 2 Raw, normalized, and ideal priority values and limit super-matrix for HSE clusters

HSE cluster	Raw score	Normalized	Ideal	Global priority
Awareness & Competency	0.358	0.358	1.000	0.358
Documentation	0.086	0.086	0.240	0.086
Emergency Preparedness	0.097	0.097	0.271	0.097
Environmental Health Management	0.093	0.093	0.260	0.093
Management-System Existence	0.148	0.148	0.413	0.148
Monitoring & Audit	0.118	0.118	0.330	0.118

To ensure the consistency of expert judgments in the ANP process, CRs were calculated for all pairwise comparison matrices using Super Decisions software. A CR threshold of 0.10 was adopted as the acceptance criterion. Matrices exceeding this threshold were returned to experts for review and revision, and iterative refinement was performed until acceptable consistency was achieved. Despite these controls, experts may encounter practical challenges during ANP pairwise comparisons, including cognitive burden due to the number of comparisons, difficulty in judging interdependent relationships, and uncertainty when comparing qualitatively different safety indicators. To mitigate these challenges, experts were provided with clear definitions of indicators, structured comparison templates, and opportunities for clarification, which helped reduce ambiguity and improve judgment coherence.

To address uncertainty and subjectivity in expert judgments, the ANP procedure was implemented using a fuzzy logic extension. Expert pairwise comparisons were expressed using triangular fuzzy numbers (TFNs), allowing linguistic preferences to be represented as bounded ranges rather than precise values. Fuzzification was performed by mapping linguistic importance scales to TFNs, and individual expert judgments were aggregated using the geometric mean to obtain consolidated fuzzy comparison matrices. Defuzzification was then conducted using the centroid method to derive crisp priorities suitable for ANP computation. All subsequent calculations, including consistency assessment

and weight derivation, were performed using Super Decisions software, which ensured methodological rigor and computational reliability without requiring manual illustration of intermediate matrices.

3.6 Empirical evaluation of contractor HSE data

Following weight derivation, the CHPI framework was empirically evaluated to assess its applicability and discriminative power across contractors. ANP-weighted indicators were used to compute composite WHSE scores, enabling benchmarking and revealing substantial variability in compliance, particularly in Emergency Preparedness and Documentation. Predictive validity was first examined using correlation and regression analyses, which showed that Awareness and Competency and WHSE Management Systems were most strongly associated with overall CHPI scores, consistent with expert prioritization. To further validate robustness, a Random Forest regression model was applied using normalized domain scores from 20 contractors. The model demonstrated strong predictive performance based on R^2, MAE, and RMSE, and feature importance rankings closely aligned with ANP-derived global weights, confirming consistency between expert judgment and data-driven validation.

Following weight derivation, the CHPI framework was empirically evaluated to assess its applicability and discriminative capacity across contractors. ANP-weighted indicators were integrated with normalized archival compliance data to compute composite WHSE scores, enabling systematic benchmarking and revealing substantial variability across safety dimensions, particularly in Emergency Preparedness and Documentation. Predictive validity was initially examined using correlation and regression analyses, which indicated that Awareness and Competency and WHSE Management Systems exhibited the strongest associations with overall CHPI scores, consistent with expert-derived prioritization. To further examine robustness and empirical coherence, a Random Forest regression model was employed as a complementary validation tool rather than a standalone predictive model.

Due to data completeness requirements, only contractors with fully available and standardized archival records were included in the machine learning analysis ($n = 20$). To mitigate overfitting risk under small-sample conditions, a 10-fold cross-validation strategy was adopted, and model performance metrics (R^2, MAE, and RMSE) were computed as averages across folds. The Random Forest model was implemented using 500 trees with bootstrap sampling enabled and mean squared error as the splitting criterion, while default depth parameters were retained to avoid excessive tuning bias. The resulting feature importance rankings demonstrated a high degree of alignment with ANP-derived global weights, supporting consistency between expert judgment and data-driven relevance. Collectively, these results confirm the empirical robustness of the CHPI framework and support the use of machine learning as a triangulation and validation mechanism for indicator prioritization rather than as a generalizable prediction engine.

4 Results

4.1 WHSE oversight mechanisms

Document analysis and field review showed that the client organization operates a multi-level oversight system regulating contractor WHSE performance across the project lifecycle. Pre-contract evaluations shortlist contractors based on documented WHSE

qualifications, followed by formal validation of appointed WHSE officers. Standardized templates and training are used to ensure consistent reporting. Monthly documentation submissions serve as operational checkpoints prior to invoice approval. Compliance is monitored through random field inspections and formal non-conformance protocols, which include financial deductions for repeated violations. At project closure, final WHSE scores are assigned using a structured form, and these scores are recorded for consideration in future contractor selection.

4.2 Derivation of core WHSE indicators

An initial set of 84 candidate WHSE performance indicators was compiled from international frameworks, sectoral guidelines, and academic literature. These indicators underwent a two-stage expert review process using the CVI method with a minimum threshold of 0.78. Items that did not meet this threshold were removed, reducing the list to 43 sub-indicators. These were grouped into seven dimensions: Awareness and Competence, Monitoring and Auditing, Emergency Response, Environmental Health Management, Documentation, Risk Assessment and Management, and Management Systems. Cronbach's alpha was calculated for the final set, with an overall reliability coefficient of 0.91 and individual dimension values ranging from 0.81 to 0.92. Tables 3 and 4 present the CVI-based reduction process and the reliability statistics for each dimension.

4.3 Weighting of indicators

To determine the relative importance of the main and sub-WHSE indicators, the ANP method was applied using structured pairwise comparisons from 14 domain experts. All computed CR were below the 0.05 threshold, confirming reliability of expert inputs. Table 5 presents the global priority weights for each main and sub-indicator. Among the seven main dimensions, Awareness and Competence achieved the highest weight (25.8%), followed by Risk Assessment and Management (21.9%) and Management Systems (14.8%). At the sub-indicator level, the highest weights were assigned to Formal WHSE Training (24.2% within its cluster), Risk Identification and Assessment Systems (21.0%), and ISO Certifications (25.1%). The lowest weights were observed in the Documentation dimension (6.7%).

4.4 Sensitivity analysis

A sensitivity analysis was performed to evaluate the effect of ± 10% perturbations in the ANP-derived weights for each of the seven WHSE clusters on the CHPI scores and rankings. For each perturbation scenario, the modified CHPI scores were recalculated while holding the remaining cluster weights constant. The results are presented in Table 6. Across all perturbations, no contractor shifted between compliance tiers. The maximum

Table 3 Content validity index (CVI) results

Evaluation step	Number of items	CVI threshold	Outcome
Initial pool (based on literature)	84	–	All items included
Post expert review (relevant + clear)	77	CVI ≥ 0.78	7 items removed after expert screening
Final CVI assessment (content validation panel)	77	CVI ≥ 0.78	34 items removed due to insufficient CVI
Final retained sub-indicators	43	–	Used in questionnaire and ANP modelling

Table 4 WHSE Main and Sub-Indicators and Cronbach's Alpha

Main indicator	Sub-indicator	Cronbach's alpha
Awareness and Competence	Official WHSE training	0.88
	WHSE Committee sessions	
	Safety briefing meetings	
	Evaluation and deployment of key WHSE personnel	
	Work performance quality	
	WHSE safety qualification certificate	
	Technical safety certification for machinery	
Monitoring and Auditing	Pre-start-up audit	0.84
	Periodic WHSE audits	
	Performance monitoring checklist completion	
	Performance & efficiency monitoring mechanism	
	Continuous monitoring of workplace pollutant levels	
Emergency Response	Emergency response plan (ERP)	0.81
	Emergency preparedness drills	
	Communication channels between client and contractor	
Environmental Health Management	Pollution control and management	0.86
	Comprehensive waste management	
	Energy consumption	
	Sanitation and hygiene conditions	
	Provision of site welfare facilities	
	Environmental management plan implementation	
	Environmental impact assessments	
Documentation	Accident and anomaly documentation	0.89
	WHSE performance checklist	
	WHSE documentation and record control	
	Occupational disease count	
	WHSE warnings follow-up	
	Occupational health exams	
	Severity rate	
Risk Assessment and Management	Risk identification and assessment system	0.92
	Safe work permit procedures	
	Corrective actions for unsafe acts/conditions	
	Essential safety equipment	
	First aid availability	
	Frequency rate	
	Safety Data Sheet (SDS/MSDS)	
	Policy statement for management systems	
	Risk control plan for unaccepted hazards	
Management Systems	ISO certifications	0.87
	WHSE organizational structure and chart	
	WHSE budget	
	Preparation of WHSE plan	

observed rank change was two positions, occurring under the Documentation cluster perturbation. Spearman's rank correlation coefficients for all scenarios remained above 0.96.

4.5 Empirical validation of the CHPI model through contractor evaluation

To verify the practical applicability of the CHPI framework, a case study evaluation was conducted on five representative contractors. Normalized performance scores were

Table 5 ANP-derived global weights and priorities

Main indicator	Main weight (%)	Sub-indicator	Sub weight (%)	Consistency ratio
Awareness & Competence	25.8	Formal WHSE Training	24.2	0.014
		Competency Evaluation & Deployment of Key WHSE Personnel	22.5	
		WHSE Committee Sessions	12.3	
		Safety Qualification Certificate	10.7	
		Technical Safety Certificate for Machinery	10.6	
		Safety Briefing Meetings	9.6	
		Work-Quality Performance	9.5	
Risk Assessment & Management	21.9	Risk Identification & Assessment System	21	0.019
		Control of acceptable HSE Risks	19.4	
		Safe Work Permit Procedures	15	
		Corrective Actions for Unsafe Acts/Conditions	14.7	
		Essential Safety Equipment	9.3	
		First Aid Availability	7	
		Safety Signage	6.8	
		Safety Data Sheet (SDS/MSDS)	6.5	
Management Systems	14.8	ISO Certifications	25.1	0.026
		WHSE Budget	24.1	
		Policy Statement for Management Systems	21.8	
		Preparation of HSE Plan	17	
		WHSE Organizational Structure & Chart	11.7	
Emergency Response	13.2	Emergency Preparedness Drills	45.3	0.035
		Emergency Response Plan (ERP)	37.4	
		Communication Channels (Client Contractor)	17.1	
Monitoring & Auditing	9.9	Pre-Start-up Audit	22.5	0.01
		Continuous Monitoring of Workplace Pollutants	17.7	
		Compliance with WHSE Regulations	16.4	
		Periodic Audits	15.8	
		Performance & Efficiency Monitoring Mechanism	15.5	
		Completion of WHSE Performance Checklist	12	
Environmental Health Management	7.7	Pollution Control & Management	18.1	0.018
		Environmental Impact Assessments	14.1	
		Comprehensive Waste Management	14.1	
		Provision of Welfare Facilities	14.1	
		Implementation of Environmental Management Plans	13.7	
		Energy Consumption	12.8	
		Sanitation & Hygiene Conditions	12.8	
Documentation	6.7	Severity Rate	20.3	0.036
		Frequency Rate	15.6	
		Occupational Health Examinations	15.3	
		Near Misses & Anomalies Documentation	13.7	
		Record Control & Documentation	12.6	
		Occupational Disease Count	11.9	
		Follow up on HSE Warnings	10.4	

assigned across the seven principal WHSE domains reflecting realistic safety maturity levels typically observed in gas infrastructure projects. Final CHPI scores were calculated through weighted summation using the ANP-derived global priorities. Based on the overall scores, contractors were classified into three compliance tiers: High (≥ 85), Moderate (75.0–84.9), and Low (< 75). As presented in Table 7, all contractors were

Table 6 Sensitivity Analysis Results for ± 10% Weight Variation

Cluster perturbed	Tier changes observed	Max rank shift	Spearman's ρ	Model stability
Awareness and Competence	0	0	0.985	Very High
Risk Assessment and Management	0	0	0.987	Very High
Management Systems	0	1	0.981	High
Emergency Response	0	1	0.978	High
Monitoring and Auditing	0	1	0.976	High
Environmental Health Management	0	1	0.968	High
Documentation	0	2	0.962	Moderate-High

placed in the Moderate tier, with scores ranging from 75.45 to 81.18. Contractor C achieved the highest CHPI score, primarily due to strong performance in Risk Assessment and Management as well as Awareness and Competence. In contrast, Contractor E obtained the lowest score, influenced by relatively lower results in Monitoring and Auditing and Emergency Preparedness.

The fact that all five contractors were classified within the "Moderate" tier should be interpreted in light of the regulatory and operational context of the studied gas infrastructure projects, where minimum WHSE compliance standards are strictly enforced, and extreme performance deviations are uncommon among active contractors. Importantly, while tier-based classification places these contractors within the same compliance category, the CHPI demonstrates meaningful discriminative capability within this tier. As shown in Table 7, CHPI scores vary by more than five points, and contractors exhibit distinct domain-level performance profiles across key dimensions such as Awareness and Competence, Risk Assessment and Management, and Monitoring and Auditing. These variations reveal substantive differences in safety maturity that would remain obscured under conventional checklist-based or threshold-driven evaluation approaches, which typically assign identical categorical ratings to contractors meeting minimum compliance requirements. In this regard, the added value of the CHPI lies not solely in categorical tier assignment, but in its ability to generate continuous, weighted performance scores that enable finer differentiation and more targeted interpretation of contractor WHSE performance, even within a single compliance tier.

4.6 Descriptive analysis of contractor data

Descriptive statistics were calculated for the seven WHSE dimensions and the composite CHPI score based on evaluations of 200 contractors. As presented in Table 8, CHPI scores ranged from 48.5 to 87.2, with a mean value of 67.87 (SD = 9.69). Across the individual dimensions, Environmental Health recorded the highest mean score (79.01), while Risk Management had the lowest (61.16). Documentation and Awareness showed the greatest dispersion, with standard deviations of 14.79 and 14.06, respectively.

4.7 Predictive modelling using random forest regression

To validate the CHPI framework's predictive potential, a Random Forest Regression model was developed using the seven core WHSE indicators as input features. The model was trained on 80% of the dataset and tested on the remaining 20%. It yielded a coefficient of determination (R^2) of 0.62, a MAE of 1.41, and a MSE of 3.33, indicating strong predictive performance with substantial explanatory power. As shown in Fig. 5, actual and predicted CHPI scores exhibit a close alignment along the identity line,

Table 7 Contractor CHPI case study results

Contractor	Awareness & competence	Risk assessment & management	Management systems	Emergency response	Monitoring & auditing	Environmental health management	Documentation	CHPI score	Tier
A	63	87	83	75	75	90	63	76.07	Moderate
B	85	67	63	78	95	86	87	78.08	Moderate
C	85	88	78	64	90	76	78	81.18	Moderate
D	73	66	93	88	83	74	89	78.55	Moderate
E	79	75	76	68	63	79	91	75.45	Moderate

particularly in the mid-to-high performance ranges, with reduced dispersion compared to the lower-performing segment. The presence of natural prediction variance, without evidence of overfitting, affirms the model's generalizability and practical relevance in contractor safety assessments.

Feature importance analysis (See Table 9) highlights Awareness & Competence (28.1%), Environmental Health (22.1%), and Monitoring & Auditing (16.8%) as the top predictors. This ranking mirrors the earlier ANP weightings and reinforces the central role of workforce capability, environmental management, and oversight in driving HSE outcomes. Across the 10-fold cross-validation procedure, model performance metrics exhibited low variability, indicating stable predictive behaviour despite the limited sample size. This consistency supports the use of Random Forest analysis as a robustness check for indicator prioritization rather than a generalizable prediction model.

4.8 Alignment between data-driven insights and ANP weights

The relative importance values obtained from the Random Forest Regression model were compared with the expert-derived global weights from the ANP process to evaluate consistency between empirical and judgment-based prioritization. As presented in Table 10, Monitoring & Auditing recorded the highest predictive importance in the ML model (37.6%) but received a lower weight in the ANP method (17.0%). Environmental Health Management ranked second in ML importance (22.5%) and displayed close agreement with its ANP weight (21.0%). Emergency Preparedness, Documentation, and Management Systems showed moderate differences between the two approaches. Risk Management and Awareness & Competence exhibited the largest negative deviations, with their ANP weights exceeding their ML-derived importance.

5 Discussion

5.1 Benchmarks with previous studies

Existing contractor safety evaluation models predominantly rely on checklist-based assessments, lagging indicators, or linear scoring systems that treat safety indicators as independent variables. Such approaches often fail to capture the interdependencies among organizational, behavioural, and operational safety factors. Moreover, prior models typically depend either on expert judgment without empirical validation, or on data-driven techniques without structured theoretical grounding. The lack of hybrid frameworks that integrate interdependency-aware weighting with predictive validation limits both the reliability and practical applicability of existing contractor safety evaluation tools. This study addresses these gaps by proposing a hybrid framework that combines fuzzy ANP to model indicator interdependencies with machine learning-based validation using real compliance data, enabling both conceptual rigor and empirical robustness in contractor safety performance assessment.

To ensure that the proposed framework is built upon a robust and conceptually valid indicator set, a systematic content validity assessment was conducted prior to weighting and modelling. Following Lawshe's content validity method, the minimum acceptable CVR was set at 0.51 for a panel of 14 experts, indicating that each retained indicator had to be judged as essential by a clear majority of the expert panel [56]. Applying this threshold directly influenced indicator selection by filtering out items with insufficient consensus and retaining only those reflecting strong domain agreement. This

Table 8 Descriptive analysis of contractor data

Factors	Mean	Std	Min	Max
Awareness	67.49	14.06	39.4	95.6
Risk Mgmt.	61.16	9.49	42.2	80.1
Emergency	71.97	8.14	55.7	88.3
Env. Health	79.01	12.21	54.6	100.0
Documentation	63.12	14.79	33.5	90.8
Monitoring	67.33	12.25	42.8	91.8
Management	63.12	13.83	35.5	90.8
CHPI	67.87	9.69	48.5	87.2

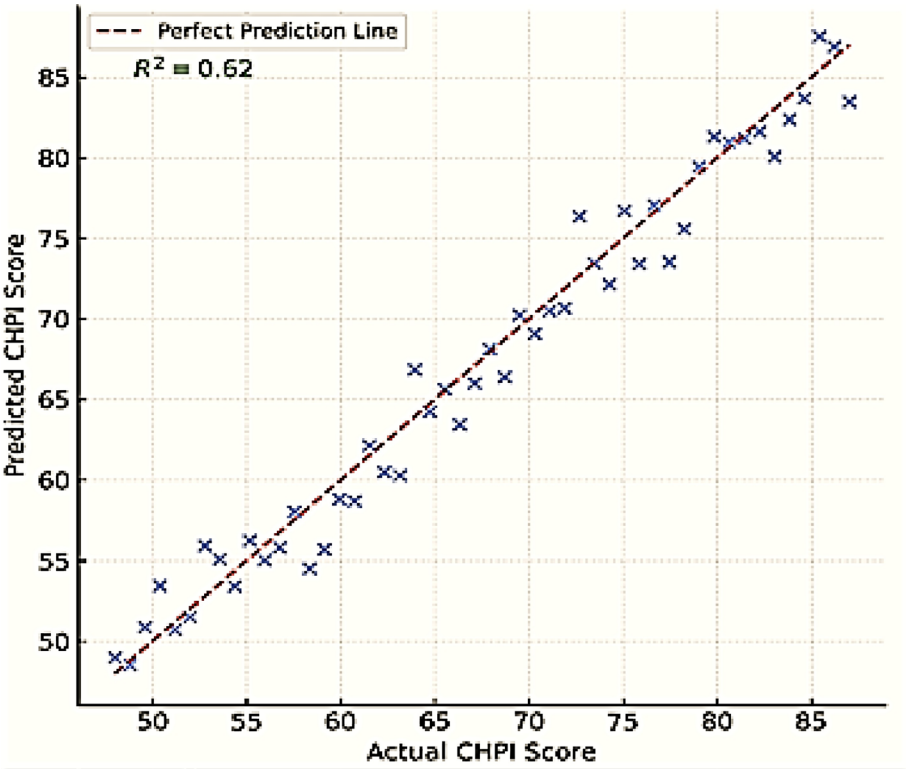


Fig. 5 Actual and predicted CHPI values

Table 9 Feature importance analysis

No.	Feature	Importance
1	Monitoring	0.1680
2	Emergency	0.1078
3	Env. Health	0.2210
4	Awareness	0.2811
5	Documentation	0.1023
6	Risk Mgmt.	0.0596
7	Management	0.0602

step reduced subjectivity in expert judgment, strengthened the theoretical grounding of the indicator pool, and ensured that subsequent ANP weighting, and machine learning validation were applied to a rigorously screened and context-relevant set of contractors WHSE indicators.

Table 10 Comparative weights of WHSE indicators from ANP and ML models

Rank	Feature	ML importance	ANP weight	Difference (ML – ANP)
1	Monitoring	0.376	0.170	+0.206
2	Env. Health	0.225	0.210	+0.015
3	Emergency	0.126	0.080	+0.046
4	Documentation	0.087	0.080	+0.007
5	Management	0.073	0.130	− 0.057
6	Risk Mgmt.	0.060	0.100	− 0.040
7	Awareness	0.053	0.230	− 0.177

The integration of ANP-derived expert weights with machine learning-based feature importance analysis in the CHPI framework represents a methodological advancement over prior contractor HSE evaluation studies, which have traditionally relied on either expert judgment or statistical correlation in isolation [57–59]. By combining a structured decision-making approach with empirical data-driven validation, this study addresses the methodological gap noted by Hill et al. [60], Tan et al. [61] and Zhang et al. [62], where the absence of a cross-verification mechanism limited the reliability of contractor performance indices. To the best of our knowledge, this is the first study to operationalize the integration of ANP-derived expert weights with machine learning-based feature importance within the CHPI framework for contractor WHSE evaluation, thereby offering a validated, dual-perspective approach that bridges conceptual priorities and empirical predictive strength.

The comparative assessment of ANP-derived weights and Random Forest Regression feature importance values revealed a mixed pattern of convergence and divergence, each carrying distinct implications for the conceptual validity and operational realism of the CHPI framework. The strongest divergence was observed in Monitoring & Auditing, which emerged as the top predictor in the ML model (37.6%) yet received a substantially lower weight in the ANP analysis (17%). This 20.6% point difference indicates that while experts recognise monitoring as a relevant dimension, they may underestimate its direct operational leverage. Empirical evidence supports the ML perspective: Gharedaghi and Omidvari [28] demonstrated that in oil and gas projects, systematic site inspections, real-time compliance tracking, and documented follow-up actions significantly outperform policy-driven measures in predicting actual HSE incident reduction. Similarly, Sattari et al. [63] found that consistent monitoring of safety-critical processes explained more variance in safety outcomes than training investments or governance structures, highlighting its high-frequency, high-impact nature. This aligns with the ML finding that monitoring provides an immediate corrective mechanism that directly influences CHPI scores. Given the limited number of contractors with complete machine-readable records, the machine learning component is intentionally positioned as a triangulation and validation mechanism rather than a definitive predictive engine.

In contrast, Environmental Health Management displayed high consistency across methods, reflecting strong conceptual and empirical agreement. Both expert judgement and data-driven modelling point to environmental safeguards as a central pillar of contractor WHSE performance. This alignment echoes findings from Amir-Heidari et al. [14] and Soleimani and Ferdos [14], who reported that environmental controls were statistically associated with higher safety compliance ratings and lower regulatory penalties in petrochemical and pipeline projects. The present results reinforce that environmental dimension serve as both a regulatory requirement and a driver of client

confidence, thereby justifying their stable weighting in integrated frameworks like CHPI. Complementary experimental evidence from construction-related environments further demonstrates that effective control of airborne pollutants, such as volatile organic compounds, plays a critical role in mitigating environmental and occupational health risks at contractor-operated sites [64]. The present results reinforce that environmental dimension serve as both a regulatory requirement and a driver of client confidence, thereby justifying their stable weighting in integrated frameworks like CHPI. Beyond routine environmental controls, evidence from safety-critical infrastructure highlights that major accident risks also emerge from structurally interacting technical, organizational, and environmental factors. Empirical structural modelling of external floating roof tanks has demonstrated that safety outcomes are shaped by interconnected system components rather than isolated control measures, reinforcing the need for interdependency-aware evaluation frameworks in contractor governance [65].

The most striking divergence in the opposite direction was seen in Awareness & Competence, which carried the highest expert derived ANP weight but ranked lowest in the ML results. This disparity suggests that while experts view training, formal qualifications, and competency validation as foundational to safety culture, these factors may exert indirect, long-term effects that are not fully captured in predictive models based on short- to medium-term outcome data. Similar patterns have been reported in other safety-related ML applications, where factors perceived by domain experts as highly influential were assigned comparatively low predictive importance by data-driven models. For instance, Zhong et al. [66] found that certain situation awareness indicators, though prioritized by pipeline safety experts, contributed minimally to ML-based risk predictions; Abbasianjahromi and Aghakarimi [67] observed comparable misalignments between managerial judgments and ML-derived feature rankings in construction safety performance; and Saridewi and Sari [68] noted that awareness and training variables in information security, while emphasized by practitioners, exhibited limited direct predictive power in ML models. Khalilzadeh et al. [69] similarly found that WHSE training initiatives improved compliance behaviours over multi-year timelines, but short-term datasets showed weak correlations because immediate operational outcomes depend more on enforcement and oversight mechanisms. Likewise, Lee et al. [42] argued that training interventions often act as enablers, enhancing the impact of other operational controls rather than producing immediate standalone effects, explaining their lower ML-derived importance in the present study.

Management Systems and Risk Management also revealed weaker predictive influence in ML analysis compared to their ANP weights. For example, Risk Management held a 10% ANP weight but only 6% ML importance, while Management Systems dropped from 13% in ANP to 7.3% in ML. Previous research by Aderamo et al. [53] and Sun et al. [70] offers a plausible explanation: governance structures, ISO certification, and risk assessment protocols create an essential enabling environment but do not directly translate into immediate performance shifts unless actively integrated into monitoring, enforcement, and operational routines. This can lead to an overestimation of their short-term impact when weights are assigned based solely on expert perception.

For Emergency Preparedness and Documentation, the ANP-ML comparison revealed relatively balanced mid-tier importance, with only modest differences. This alignment suggests that these domains act as bridging factors, valued by experts for their role in

structural readiness while also demonstrating measurable operational performance benefits in ML models. Similar findings are reflected in prior research. Kyrkou et al. [71], Phark et al. [72], Richardson [73], and Damaševičius et al. [74] collectively show that emergency planning and preparedness not only provide an organizational safety backbone but also yield quantifiable improvements in response time and risk mitigation when integrated with machine learning and sensor-based systems. Likewise, in the domain of safety documentation, Khan et al. [75], Fenton and Simske [76], and Kumi et al. [40] demonstrate that intelligent document processing and digitalized record-keeping enhance both compliance and the efficiency of safety management workflows, aligning with the dual structural-operational role observed in our results.

Overall, the combination of ANP and ML results highlights the need for a dual-perspective weighting strategy in future CHPI iterations. ANP effectively captures long-term, culture-building priorities as perceived by domain experts, while ML uncovers high-frequency, high-impact levers that directly influence measurable safety outcomes. As Trivedi et al. [77] recommend, hybrid weighting systems that reconcile conceptual importance with empirical predictive validity can yield more balanced and context-responsive performance indices. In practice, this means retaining the structural emphasis on training, management systems, and risk protocols while increasing the operational weight of monitoring and environmental controls to reflect their proven predictive strength. By explicitly unpacking these convergences and divergences, the CHPI framework offers both a validated measurement tool and a diagnostic lens to reconcile expert judgment with operational data, enhancing transparency, policy relevance, and transferability to high-risk industries beyond the gas sector.

5.2 Reconciling expert-based and data-driven weights in future CHPI development

The observed discrepancies between ANP-derived weights and machine learning-based feature importance values carry important implications for the future design and refinement of the CHPI framework. While the ANP prioritization reflects expert judgment regarding long-term structural and cultural drivers of contractor WHSE performance, the Random Forest results emphasize operational indicators with more immediate and measurable influence on empirical performance outcomes. These differences highlight that expert-based and data-driven weighting approaches capture complementary, rather than conflicting, dimensions of safety performance. From an index design perspective, this divergence suggests that relying exclusively on either expert judgment or machine learning may lead to partial representations of contractor safety performance. Expert-based weighting is well suited for capturing latent, system-level factors such as competency development, governance structures, and safety culture, whose effects may unfold over longer time horizons. In contrast, data-driven models tend to prioritize high-frequency, enforcement-oriented indicators, such as monitoring, auditing, and environmental controls, that exert direct and short-term influence on observed compliance outcomes.

Future versions of the CHPI could address this methodological tension through the development of a hybrid weighting system that integrates expert-derived and data-driven weights in a structured manner. One possible approach involves combining normalized ANP weights and machine learning importance scores using adjustable blending coefficients, allowing decision-makers to balance strategic, long-term priorities

against operational performance sensitivity. Alternatively, expert weights may be treated as baseline structural weights, while machine learning outputs are used as dynamic adjustment factors that respond to evolving performance data. Such hybrid weighting mechanisms would preserve the conceptual integrity and interpretability of the CHPI while enhancing its empirical responsiveness and adaptability. By explicitly integrating both judgment-based and data-driven perspectives, future CHPI implementations could achieve a more balanced, context-aware CHPI capable of supporting both strategic safety governance and real-time contractor performance management.

5.3 Theoretical and methodological contributions

This study contributes meaningfully to both the theoretical discourse and methodological advancement of WHSE performance evaluation. Theoretically, it reinforces and operationalizes the systems thinking paradigm within safety science, particularly in contractor governance. By modelling contractor safety performance as a complex, interdependent network of behavioural, organizational, and procedural factors, the CHPI framework aligns with socio-technical systems theory such as Dahl and Kongsvik [78] and Lingard and Pirzadeh [79]. Rather than treating safety as the outcome of isolated activities, the framework emphasizes how the effectiveness of each domain, such as emergency preparedness or documentation, depends on the strength of others like competence or audit mechanisms. This holistic view advances prior work that typically approached safety indicators as independent variables. The emphasis on interactions, captured through the ANP, mirrors arguments from Salas and Hallowell [48], who noted that safety leading indicators are only meaningful when situated within broader systemic functions, such as communication quality, behavioural adherence, and feedback structures.

Methodologically, the integration of a three-stage framework, Delphi consensus, CVI filtering, and ANP modelling, offers a novel structure for developing and validating performance metrics. While Delphi panels and CVI have been used separately in occupational safety studies such as Rshidi et al. [80], their combination with network-based weighting introduces a level of methodological rigor previously absent from contractor-specific frameworks. The result is an indicator set that is not only psychometrically validated but also hierarchically interlinked, enabling more realistic prioritization. In comparison to earlier studies, such as Ahmed [81] or Nassiri et al. [82], which relied on binary or frequency-based scoring, the CHPI model offers graded, empirically ranked prioritization. It further introduces the ability to model reciprocal feedback, studies like Hadidi and Khater [83] could not address.

Integrating real audit data anchors, the CHPI model in practice, transforming it into an operational decision-support tool. Random Forest regression provides data-driven validation, while ANP-ML comparisons reveal both alignment and variation, demonstrating analytical depth. Mapping the ANP structure to Safety Performance Theory and Socio-Technical Systems Theory further strengthens its validity and bridges theoretical constructs with practical WHSE evaluation (Fig. 6).

Building on these contributions, the proposed framework also offers a clear analytical distinction between operational performance drivers and structural safety controls in contractor WHSE evaluation. The ANP-based expert weighting primarily reflects structural controls, such as management systems, competency development, and formal

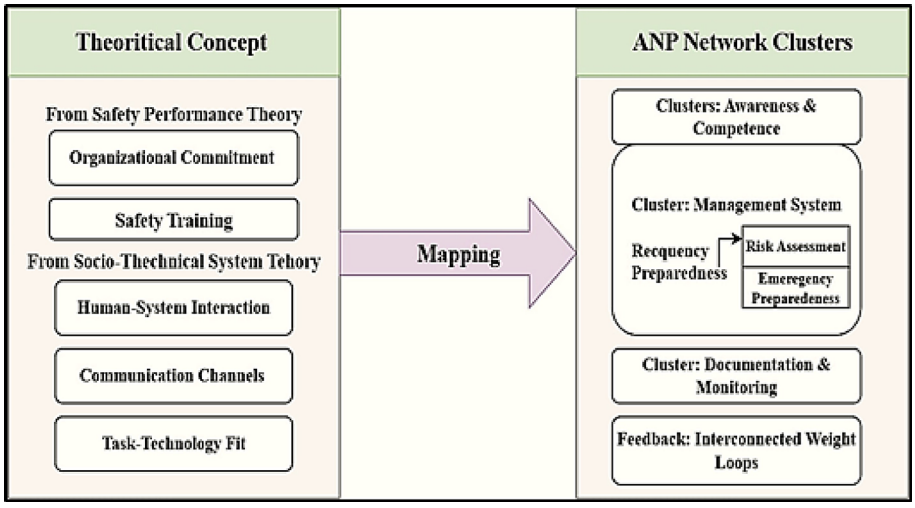


Fig. 6 Theoretical-to-operational mapping of the CHPI framework

risk governance mechanisms, which shape the long-term safety capacity of contractor organizations. In contrast, the machine learning-based feature importance analysis highlights operational performance drivers, including monitoring, auditing, and environmental health practices, that exert immediate and observable influence on WHSE outcomes. By integrating these two analytical perspectives within a unified framework, the study demonstrates how short-term operational levers and long-term structural controls contribute differently, but complementarily, to contractor safety performance. This distinction provides both theoretical clarity and practical guidance that is largely absent from prior contractor safety evaluation models relying on single-method approaches.

Beyond this analytical distinction, the added value of the CHPI lies in its ability to overcome key limitations of existing contractor safety evaluation methods. Unlike conventional checklist-based or scorecard approaches that generate static and weakly discriminative safety scores, the CHPI produces an empirically grounded performance index that reflects both structural dependencies and operational influence. Compared to prior multi-criteria models that rely solely on expert judgment, the inclusion of machine learning-based validation enhances the credibility and decision relevance of the results by linking indicator importance to observed performance outcomes. At the same time, in contrast to purely data-driven safety models that prioritize prediction at the expense of interpretability, the CHPI preserves transparency by embedding predictive insights within a theory-informed indicator structure. Collectively, these features position the CHPI as an integrative and scalable advancement that provides clearer prioritization, stronger justification for intervention, and greater practical utility than existing contractor safety evaluation approaches.

5.4 Practical and policy implications

The CHPI framework explicitly bridges the gap between operational contractor performance evaluation and established theoretical safety frameworks by translating abstract safety concepts into measurable, decision-relevant indicators. While theories such as Safety Performance Theory and Socio-Technical Systems Theory emphasize the systemic, interdependent, and dynamic nature of safety, their application in contractor evaluation has often remained conceptual. The CHPI operationalizes these

theoretical principles by embedding them within a structured multi-criteria decision-making model, where expert-informed interdependencies are quantified through ANP and empirically examined using real compliance data and machine learning validation. In doing so, the CHPI connects theoretical understanding of safety as an emergent system property with practical, field-level contractor assessment, enabling theory-informed yet operationally actionable safety governance.

The development of the CHPI represents not only a practical decision-support tool but also a core theoretical and methodological contribution of this study. Unlike conventional safety scorecards or checklist-based evaluations, the CHPI consolidates interdependent behavioural, organizational, and procedural indicators into a unified, empirically weighted structure that reflects the systemic nature of contractor performance. By embedding validated indicators within a Fuzzy ANP network and calibrating their weights using real audit data, the CHPI operationalizes safety performance as an emergent property of interacting subsystems rather than discrete metrics. This design captures the latent interconnections that underpin safety outcomes and transforms fragmented safety data into a coherent composite index, enabling robust differentiation among contractors with varying maturity levels. In doing so, the CHPI advances existing safety performance models by shifting the focus from compliance-driven measurement toward intelligence-driven decision support.

Building on this foundation, the CHPI offers significant implications for organizational safety management and policy development in high-risk industries. Practically, it provides a structured, evidence-based framework for improving decision-making across all phases of contractor engagement—supporting pre-contract screening, ongoing audits, and post-project evaluations. It enables more efficient resource allocation by identifying high-risk contractors and tailoring interventions such as training or supervision accordingly. ANP-derived weightings can guide internal budgeting, encouraging a shift from passive compliance checks toward proactive, behaviour-based safety management. In multi-contractor environments, the CHPI also facilitates performance benchmarking and supports the design of incentive or penalty schemes.

At the policy level, the CHPI presents a scalable framework for performance-based regulation. Its indicators can be adopted by regulatory bodies to establish objective safety criteria in licensing and accreditation, particularly in contexts with limited enforcement capacity. The model's modular design allows integration into digital procurement platforms and predictive analytics systems, enhancing real-time oversight and early intervention. Beyond the gas sector, the CHPI's modular architecture allows the indicator pool to be reconfigured according to sector-specific hazards while preserving the underlying multi-criteria decision structure, making it suitable for domains such as petrochemicals, mining, construction, and transportation. Broader adoption of the CHPI can therefore promote consistency in safety standards across sectors, fostering transparency and strengthening trust among stakeholders. By demonstrating how a rigorously validated fuzzy-ML hybrid approach can produce a scalable CHPI, this study offers a blueprint for extending the CHPI beyond contractor safety assessment toward broader applications in organizational risk governance and performance analytics.

More specifically, the CHPI provides a practical foundation for advancing performance-based regulation in the oil and gas sector, where traditional prescriptive compliance approaches have shown limited effectiveness in managing contractor-related risks.

By translating multi-dimensional WHSE performance into a transparent, composite index grounded in both expert judgment and empirical data, the CHPI enables regulators and client organizations to shift from rule-based oversight toward outcome-oriented regulation. Regulatory bodies can use CHPI scores to inform risk-based licensing, differentiated inspection frequencies, and performance-linked incentives or penalties for contractors. This approach supports proportional regulatory responses, whereby high-performing contractors are rewarded with reduced administrative burden, while lower-performing contractors are subject to intensified oversight and targeted corrective actions. In this way, the CHPI operationalizes performance-based regulation by aligning regulatory intervention with demonstrable safety outcomes, enhancing accountability, and promoting continuous improvement in contractor WHSE performance within the oil and gas industry.

5.5 Limitations and future research directions

Despite its strong theoretical and practical foundation, the CHPI model has several limitations. The ANP weighting relied on input from 14 experts, which, while experienced, may limit generalizability across industries and regions. Additionally, some compliance records lacked full documentation, requiring inferential scoring that could introduce minor uncertainties. Future research should test the CHPI in other high-risk sectors like petrochemicals, mining, and large infrastructure projects to assess its transferability. Longitudinal studies using time-series data are also needed to evaluate its predictive accuracy over time. Further development could involve integrating advanced machine learning to dynamically adjust weights and detect real-time anomalies. Creating interactive dashboards to monitor contractor risk and compliance trends would enhance practical utility.

A further limitation relates to the machine learning validation component. Although the overall CHPI evaluation covered a larger contractor pool, only a subset of contractors had complete, standardized archival records suitable for Random Forest analysis. While the use of 10-fold cross-validation mitigates overfitting risk, the ML results should be interpreted as supporting evidence for indicator relevance rather than as a fully generalizable predictive model. Future studies with larger longitudinal datasets could enable more advanced tuning and uncertainty estimation.

While the proposed CHPI framework demonstrates strong methodological rigor and practical relevance, several considerations should be acknowledged to contextualize its applicability. First, the model was developed and empirically validated within the oil and gas sector, where contractor operations involve high-risk processes, strict regulatory requirements, and complex socio-technical interactions. Although the core structure of the CHPI is grounded in general safety science principles, the relative importance of specific indicators may vary across industries with different hazard profiles or organizational arrangements. Second, expert judgments were elicited from a purposively selected panel with extensive domain experience, which supports internal validity but may limit generalizability to contexts with differing regulatory or cultural conditions. These considerations do not undermine the robustness of the proposed framework but rather highlight opportunities for future enhancement. Subsequent research may extend validation of the CHPI across other high-risk sectors, such as petrochemicals, construction, or mining, to examine the stability of indicator interdependencies under varying

operational conditions. Future studies may also explore adaptive or dynamic weighting mechanisms that update indicator priorities in response to real-time performance data, as well as integration with digital safety dashboards or predictive risk management platforms. Such extensions would further strengthen the scalability and long-term applicability of the CHPI as a flexible tool for performance-based safety governance.

6 Conclusion

This study developed and validated the CHPI, a novel multi-criteria framework for evaluating contractor safety performance in high-risk industrial sectors. By integrating expert driven fuzzy ANP weightings with empirical compliance data and machine learning validation, the CHPI moves beyond traditional checklist or scorecard approaches. It captures the interdependent structure of safety indicators and links them with real-world performance data, operationalizing safety as a systemic property rather than isolated metrics. Empirical analysis revealed that indicators related to Monitoring and Environmental Health Management exerted the strongest influence on overall WHSE performance, and the strong alignment between ANP-derived weights and Random Forest feature importance confirmed the internal validity and conceptual soundness of the CHPI model. Predictive testing further showed that CHPI scores can reliably distinguish between contractors with differing safety risk profiles, demonstrating its practical value as a proactive decision-support tool.

Beyond its methodological contribution, the CHPI carries significant implications for policy and practice. It enables more strategic allocation of oversight resources, supports performance-based contractor selection, and offers a foundation for integration into digital procurement and compliance platforms. Its modular architecture allows adaptation across sectors and regulatory environments, contributing to the harmonization of safety standards and strengthening accountability in contractor governance. Overall, the CHPI represents a step change in contractor safety evaluation, bridging the gap between theoretical rigor and operational applicability. Future research should extend its validation across diverse industries, incorporate adaptive weighting mechanisms responsive to dynamic risk conditions, and develop real-time dashboard applications to support continuous safety intelligence. As safety challenges grow in complexity, tools like CHPI provide a pathway toward more intelligent, responsive, and accountable risk management in the contractor ecosystem.

From an applied standpoint, the results suggest that contractor safety governance should move beyond predominantly documentation-oriented evaluations toward more dynamic, performance-sensitive oversight strategies. In particular, the dominant influence of Monitoring and Environmental Health Management highlights the importance of continuous site inspections, real-time compliance tracking, and timely corrective actions as primary levers for improving WHSE outcomes. Client organizations and regulators can operationalize the CHPI by embedding it within contractor prequalification systems, periodic performance audits, and incentive-based management schemes, thereby enabling targeted interventions for high-risk contractors. Moreover, the observed divergence between expert-derived priorities and machine learning importance underscores the need for balanced safety strategies that integrate long-term capacity-building measures with short-term operational controls. By translating analytical findings into concrete governance mechanisms, the CHPI provides a practical pathway

for strengthening accountability, enhancing transparency, and supporting evidence-based decision-making in complex contractor environments.

Quantitatively, the proposed CHPI framework demonstrated strong discriminatory and predictive capability. Indicators associated with Monitoring and Environmental Health Management consistently ranked among the top-weighted dimensions, accounting for the largest share of influence on overall WHSE performance. The Random Forest model achieved robust predictive performance, with CHPI scores explaining a substantial proportion of variance in contractor safety outcomes and enabling clear differentiation between higher- and lower-risk contractors. Importantly, the convergence between expert-derived ANP weights and data-driven feature importance provides quantitative confirmation that the identified priority indicators are not only conceptually meaningful but also empirically relevant. These results substantiate the practical utility of the CHPI as a measurable, evidence-based tool for performance-oriented contractor safety management in high-risk industrial settings.

Acknowledgements
The authors gratefully acknowledge the West Azerbaijan Gas Company for their support during data collection and Tabriz University of Medical Sciences for their academic and logistical assistance.

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Funding
This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
for this study was obtained from the Ethics Committee of Tabriz University of Medical Sciences (IR.TBZMED.REC.1398.416) and the Research Coordination Office of EAGC. All participants provided informed consent to participate and agreed to the publication of anonymized data. The study procedures complied with the ethical standards of the institutional research committee and the principles of the Helsinki Declaration. All expert participants involved in the content validation and ANP weighting phases were informed about the study objectives, procedures, and intended use of the findings prior to participation. Participation was entirely voluntary, and written informed consent was obtained from all experts. For the archival contractor compliance data, access was granted through formal authorization by the host organization, and no personally identifiable information was collected or analysed. All data were handled confidentially and used exclusively for research purposes.

Consent for publication
All authors have reviewed and approved the final version of the manuscript and consent to its publication. The participating experts provided consent for the use and publication of anonymized findings derived from their evaluations. The host organization granted formal permission for the use and publication of aggregated contractor compliance data. No identifiable personal or organizational information is disclosed in this manuscript.

Clinical trial number
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 19 January 2026 / Accepted: 3 March 2026
Published online: 28 April 2026

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